Math 477 – Midterm 1 Study Guide

Counting, Axioms of Probability, Conditional Probability & Independence, Discrete Random Variables

Compiled from Prof. Salamon's lecture notes, the Midterm 1 review sheet, and Ross, *A First Course in Probability* (10th ed.)

Exam: Thursday, October 15, 2026

How to use this guide

Scope: everything through Sec. 4.5. **Not tested:** 1.6, 2.6, 2.7, 3.5, proof by induction, and everything from 4.6 on (binomial, Poisson, ...).

Colors: blue = definition, red = theorem/formula, green = worked example, orange = common mistake.

Sources: examples marked *(lecture)* are from the professor's notes; *(Ross Ex. xx)* are from the textbook. Section numbers match the textbook's.

Professor's priority order: (1) the review sheet – Section 5 is a checklist and Section 6 has worked solutions; (2) the textbook's **Self-Test Problems and Exercises** – Section 7 lists which ones are in scope; (3) hand-in homework; (4) lecture examples; (5)–(6) other problems and odd textbook examples.

Reminder: definitions must be stated precisely and unambiguously – grammar and spelling are not graded, but clarity is.

Contents

1 Chapter 1 – Combinatorial Analysis **3**

1.1 The Basic Principle of Counting (Sec. 1.2) . . . 3
1.2 Permutations (Sec. 1.3) . . . 4
1.3 Combinations (Sec. 1.4) . . . 5
1.4 The Binomial Theorem (Sec. 1.4) . . . 6
1.5 Partitions / Multinomial Coefficients (Sec. 1.5) . . . 6

2 Chapter 2 – Axioms of Probability **7**

2.1 Sample Space and Events (Sec. 2.2) . . . 7
2.2 Axioms of Probability (Sec. 2.3) . . . 8
2.3 Some Simple Propositions (Sec. 2.4) . . . 8
2.4 Sample Spaces with Equally Likely Outcomes (Sec. 2.5) . . . 9

3 Chapter 3 – Conditional Probability and Independence **11**
3.1 Conditional Probability (Sec. 3.2) . . . 11
3.2 Bayes's Formula (Sec. 3.3) . . . 12
3.3 Independent Events (Sec. 3.4) . . . 14

4 Chapter 4 – Discrete Random Variables (4.1–4.5) **15**
4.1 Random Variables, PMF, CDF (Sec. 4.1) . . . 15
4.2 Discrete Random Variables (Sec. 4.2) . . . 16
4.3 Expected Value (Sec. 4.3) . . . 17
4.4 Expectation of a Function of a R.V. (Sec. 4.4) . . . 17
4.5 Variance (Sec. 4.5) . . . 18

5 Review-Sheet Checklist (state, formula, explain, example) **19**

6 Review Sheet Practice Problems – Worked Solutions **19**

7 Textbook Self-Test Problems – What's In Scope **20**

8 Quick-Reference Formula Sheet **21**

1 Chapter 1 – Combinatorial Analysis

Why counting? Many probability problems reduce to counting: probability = (number of favorable outcomes)/(number of outcomes) when outcomes are equally likely. (Ross opens with the antenna problem: n antennas in a row with m defective; the system works if no two defectives are adjacent; e.g. $n = 4, m = 2$ gives 3 working arrangements out of 6.)

1.1 The Basic Principle of Counting (Sec. 1.2)

Basic Principle of Counting / Multiplication Principle

If experiment 1 has m possible outcomes and, for *each* of them, experiment 2 has n possible outcomes, then together there are mn outcomes. In lecture notation, for stages A and B: $N(S) = N(A) \cdot N(B)$.

Generalized principle: for r stages where stage i has n_i outcomes (regardless of earlier results),

$$N(S) = n_1\, n_2 \cdots n_r.$$

Proof idea: list the outcomes as an $m \times n$ array of pairs (i, j): m rows of n entries.

Examples

- *(Ross Ex. 2a)* 10 women, each with 3 children; choose a mother and one of her children: $10 \cdot 3 = 30$.
- *(Ross Ex. 2b)* 3 freshmen, 4 sophomores, 5 juniors, 2 seniors; subcommittee with one from each class: $3 \cdot 4 \cdot 5 \cdot 2 = 120$.
- *(Ross Ex. 2c)* 7-place plates, first 3 letters then 4 digits: $26^3 \cdot 10^4 = 175{,}760{,}000$. *(Ross Ex. 2e)* With no repetition: $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 \cdot 7 = 78{,}624{,}000$.
- *(Ross Ex. 2d)* Functions on n points with values in $\{0, 1\}$: 2^n.
- *(lecture)* Coin and die: $2 \cdot 6 = 12$; coin and two dice: 72. Plates (3 digits, then 4 letters): $10^3 \cdot 26^4 = 456{,}976{,}000$. If 7 four-letter words are prohibited: $10^3(26^4 - 7) = 456{,}969{,}000$ (**not** $10^3 \cdot 26^4 - 7$).
- *(Review #2)* 7-digit phone numbers, first digit not 0 or 1, no 555-xxxx: $8 \cdot 10^6 - 10^4 = 7{,}990{,}000$.

1.2 Permutations (Sec. 1.3)

Factorial and Permutation

$n! = 1 \cdot 2 \cdot 3 \cdots (n-1) \cdot n$, with $0! = 1$ and $n! = n\,(n-1)!$.

A **permutation** is an *ordered* arrangement/selection of objects.

- Orderings of n distinguishable objects: $n!$ (first slot n choices, then $n-1$, ...).
- Ordered selections of r from n distinguishable objects: ${}_nP_r = \dfrac{n!}{(n-r)!} = n(n-1)\cdots(n-r+1)$. (Note ${}_nP_n = n!/0! = n!$.)

Permutations of n objects with indistinguishable ones

If n_1 of the objects are alike, n_2 are alike, ..., n_r are alike ($n_1 + \cdots + n_r = n$), the number of distinct arrangements is

$$\frac{n!}{n_1!\,n_2!\cdots n_r!}.$$

Why: the $n!$ labelled orderings overcount each pattern by $n_1!\,n_2!\cdots n_r!$ (permuting alike objects among themselves doesn't change the pattern).

Examples

- *(Ross Ex. 3a)* Batting orders for 9 players: $9! = 362{,}880$.
- *(Ross Ex. 3b)* 6 men and 4 women ranked on an exam: $10! = 3{,}628{,}800$ rankings; if men are ranked among themselves and women among themselves: $6!\,4! = 17{,}280$.
- *(Ross Ex. 3c)* 10 books (4 math, 3 chemistry, 2 history, 1 language), same subject together: $4! \cdot (4!\,3!\,2!\,1!) = 6912$ (order the subjects, then order within each subject).
- *(Ross Ex. 3d)* Arrangements of PEPPER: $\frac{6!}{3!\,2!} = 60$. *(lecture)* MISSISSIPPI: $\frac{11!}{4!\,4!\,2!\,1!} = 34{,}650$.
- *(Ross Ex. 3e)* Chess tournament, 10 players (4 Russian, 3 US, 2 British, 1 Brazilian), list only nationalities by finish: $\frac{10!}{4!\,3!\,2!\,1!} = 12{,}600$.
- *(Ross Ex. 3f)* Signals of 9 flags from 4 white, 3 red, 2 blue identical flags: $\frac{9!}{4!\,3!\,2!} = 1260$.
- *(lecture)* Photo of 4 of 10 people in a line: ${}_{10}P_4 = 5040$. *(Review #3)* 12 blocks (4 red, 6 black, 1 white, 1 blue): $\frac{12!}{4!\,6!\,1!\,1!} = 27{,}720$.

1.3 Combinations (Sec. 1.4)

Combination

An *unordered* selection of r objects from n objects (a subset of size r):

$$\binom{n}{r} = {}_nC_r = \frac{n!}{(n-r)!\,r!} \qquad (\text{“}n \text{ choose } r\text{”}).$$

Convention: $\binom{n}{r} = 0$ if $r < 0$ or $r > n$. Also $\binom{n}{0} = \binom{n}{n} = 1$ (one empty subset, one full subset).
Why: ${}_nP_r$ counts ordered selections; each group of r is counted $r!$ times, so $\binom{n}{r} = {}_nP_r/r!$.
Symmetry: $\binom{n}{r} = \binom{n}{n-r}$ (choosing the chosen = choosing the left-out).

Pascal's identity (Ross eq. 4.1)

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}, \qquad 1 \le r \le n.$$

Combinatorial proof: fix object 1. Groups of size r that contain object 1: $\binom{n-1}{r-1}$; groups that do not: $\binom{n-1}{r}$.

Examples

- *(Ross Ex. 4a)* Committee of 3 from 20 people: $\binom{20}{3} = 1140$. *(lecture)* 3 from 9: $\binom{9}{3} = 84 = \binom{9}{6}$.
- *(Ross Ex. 4b / lecture)* 5 women, 7 men; committees of 2 women and 3 men: $\binom{5}{2}\binom{7}{3} = 350$. If 2 men are feuding and won't serve together: committees with both feuders are $\binom{5}{2} \cdot \binom{5}{1}$ (the pair plus 1 of the other 5 men), so

$$\binom{5}{2}\left[\binom{7}{3} - \binom{5}{1}\right] = 10\,(35-5) = 300.$$

- *(Ross Ex. 4c)* n antennas, m defective (indistinguishable), no two defectives adjacent: line up the $n-m$ working ones and choose m of the $n-m+1$ gaps: $\binom{n-m+1}{m}$.
- *(Ross Ex. 4e)* Number of subsets of an n-element set: $\sum_k \binom{n}{k} = 2^n$ (each element in or out). Nonempty subsets: $2^n - 1$.

Choosing the right tool

	no replacement	**with replacement**
order matters	${}_nP_r$	n^r (multiplication principle)
order doesn't matter	$\binom{n}{r}$	(harder – not needed)

Distinguishable objects into distinguishable slots → permutations. Indistinguishable objects chosen from distinguishable ones (no order) → combinations. *Subtraction trick:* count everything, then subtract the forbidden cases (feuding men).

1.4 The Binomial Theorem (Sec. 1.4)

Binomial Theorem (Ross eq. 4.2)

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k\, y^{n-k}.$$

Combinatorial proof: $(x_1+y_1)(x_2+y_2)\cdots(x_n+y_n)$ expands into 2^n terms, each choosing x_i or y_i from every factor. The terms with exactly k x's correspond to choosing which k of the n factors give an x: $\binom{n}{k}$ terms. Set all $x_i = x$, $y_i = y$. (Ross also gives an induction proof via Pascal's identity – induction is not tested.)
Examples: $(x+y)^3 = y^3 + 3xy^2 + 3x^2y + x^3$. $\sum_k \binom{n}{k} = 2^n$ (set $x = y = 1$); $\sum_k (-1)^k \binom{n}{k} = 0$ (set $x = 1, y = -1$). Coefficients are the rows of **Pascal's triangle**: 1; 1 1; 1 2 1; 1 3 3 1; 1 4 6 4 1; ...

1.5 Partitions / Multinomial Coefficients (Sec. 1.5)

Partition

Let A be a set of n distinct objects and let $n_1, \ldots, n_r$ be whole numbers summing to n. A distribution of the n objects into (distinct) groups of sizes $n_1, \ldots, n_r$ is a **partition** of A. The number of partitions is the **multinomial coefficient**

$$\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1!\, n_2! \cdots n_r!}.$$

A combination is the special case $r = 2$.

Proof: choose group 1, then group 2 from the rest, and so on: $\binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\cdots-n_{r-1}}{n_r}$; factorials telescope to $\frac{n!}{n_1!\cdots n_r!}$. *Second view:* it equals the number of arrangements of n symbols of which n_i are alike – so "letters in a word" and "people in committees" are the same computation (slots are the people; letters are the committees).

Multinomial Theorem

$$(x_1 + x_2 + \cdots + x_r)^n = \sum_{n_1+\cdots+n_r=n} \binom{n}{n_1, \ldots, n_r} x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r}.$$

(e.g. $(x_1+x_2+x_3)^2 = x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 + 2x_1x_3 + 2x_2x_3$.)

Examples

- *(lecture)* 9 people into committees of sizes 2, 3, 4: $\frac{9!}{2!\,3!\,4!} = 1260$.
- *(Ross Ex. 5a)* 10 police officers: 5 patrol, 2 at the station, 3 reserve: $\frac{10!}{5!\,2!\,3!} = 2520$.
- *(Ross Ex. 5b)* 10 children into an A team and a B team of 5 each: $\frac{10!}{5!\,5!} = 252$.
- *(Ross Ex. 5c)* 10 children split into two teams of 5 with **no labels** (order of teams irrelevant): $\frac{10!/(5!\,5!)}{2!} = 126$.

Watch out

Labeled groups (an A team and a B team) $\to$ multinomial. **Unlabeled groups of equal size** $\to$ divide by the number of ways to reorder the groups (2! above). Sec. 1.6 (integer solutions of $x_1 + \cdots + x_r = n$) is *not* tested.

2 Chapter 2 – Axioms of Probability

2.1 Sample Space and Events (Sec. 2.2)

Sample space, event

The **sample space** S is the set of all possible outcomes of an experiment. An **event** is a subset of the sample space, $E \subseteq S$. We say E *occurs* if the outcome lies in E.
Ross examples: sex of a newborn $S = \{g, b\}$; order of finish of 7 horses $S =$ all 7! permutations; two coins $S = \{(h,h),(h,t),(t,h),(t,t)\}$; two dice $S = \{(i,j) : 1 \le i,j \le 6\}$ (36 points); lifetime of a transistor $S = \{x : 0 \le x < \infty\}$.
Events: "at least one head" $= \{(h,h),(h,t),(t,h)\}$; "sum of dice is 7" $= \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$.

Set notation

$x \in A$	member of	$\emptyset$	empty (null) set/event
$A \subseteq B$	every element of A is in B	$N(A)$ or $n(A)$	number of elements
$A \cup B$	in A *or* B (or both)	$A \cap B = AB$	in A *and* B
$A^c = A'$	complement (in S, not in A)	$A - B$	in A but not in B

$\bigcup_{i=1}^n E_i$ (at least one occurs) and $\bigcap_{i=1}^n E_i$ (all occur). Note $S^c = \emptyset$. If $E \subseteq F$, then the occurrence of E implies the occurrence of F. U is the universal set ("know your universe").

Mutually exclusive events

A and B are **mutually exclusive** (disjoint) if $A \cap B = \emptyset$: no outcome lies in both, so they cannot both occur. Example: "sum is 7" and "sum is 6" for two dice.

Set laws

De Morgan: $\left(\bigcup_i E_i\right)^c = \bigcap_i E_i^c$, $\left(\bigcap_i E_i\right)^c = \bigcup_i E_i^c$; for two events $(A \cup B)^c = A^c B^c$ and $(AB)^c = A^c \cup B^c$.
Commutative: $E \cup F = F \cup E$, $EF = FE$.
Associative: $(E \cup F) \cup G = E \cup (F \cup G)$, $(EF)G = E(FG)$.
Distributive: $(E \cup F)G = EG \cup FG$, $(EF) \cup G = (E \cup G)(F \cup G)$.
(Note $(E \cup F) \cap G \neq E \cup (F \cap G)$ in general.) Verify any of these with Venn diagrams: an outcome is in the left side iff it is in the right.

2.2 Axioms of Probability (Sec. 2.3)

The 3 Axioms

A probability P assigns to each event E a number $P(E)$ such that

1. $0 \le P(E) \le 1$,
2. $P(S) = 1$,
3. for any sequence of mutually exclusive events $E_1, E_2, \ldots$: $P(\bigcup_i E_i) = \sum_i P(E_i)$.

The *relative frequency* definition $P(E) = \lim_{n\to\infty} n(E)/n$ has the drawback that the limit's existence must be assumed; the axiomatic approach avoids this (and the strong law of large numbers shows the limit exists with probability 1).

First consequences

$P(\emptyset) = 0$ (take $E_1 = S$, $E_i = \emptyset$ for $i > 1$ in Axiom 3: $P(S) = P(S) + \sum P(\emptyset)$).
Finite additivity: for mutually exclusive $E_1, \ldots, E_n$, $P(\bigcup E_i) = \sum P(E_i)$.

Examples

- *(lecture)* Fair coin: H, T mutually exclusive, $P(H) = P(T)$, $1 = P(S) = P(H) + P(T) = 2P(H) \Rightarrow P(H) = \frac{1}{2}$.
- *(Ross Ex. 3a)* Biased coin where heads is twice as likely as tails: $P(H) = \frac{2}{3}$, $P(T) = \frac{1}{3}$.
- *(Ross Ex. 3b)* Fair die: $P(\text{even}) = P(2) + P(4) + P(6) = \frac{1}{2}$ by Axiom 3.

2.3 Some Simple Propositions (Sec. 2.4)

Propositions 4.1–4.4

1. **Complement rule:** $P(E^c) = 1 - P(E)$. *Proof:* E, E^c are mutually exclusive with union S: $1 = P(E) + P(E^c)$.
2. **Monotonicity:** if $E \subseteq F$ then $P(E) \le P(F)$. *Proof:* $F = E \cup E^c F$ (disjoint), so $P(F) = P(E) + P(E^c F) \ge P(E)$.
3. **Union of two events:** $P(E \cup F) = P(E) + P(F) - P(EF)$. *Proof:* $E \cup F = E \cup E^c F$ (disjoint) and $P(E^c F) = P(F) - P(EF)$.
4. **Inclusion–exclusion (3 events):**

$$\begin{aligned} P(E \cup F \cup G) = {} & P(E) + P(F) + P(G) \\ & - P(EF) - P(EG) - P(FG) + P(EFG). \end{aligned}$$

5. **Inclusion–exclusion (general):**

$$\begin{aligned} P\Big(\bigcup_{i=1}^{n} E_i\Big) = {} & \sum_i P(E_i) - \sum_{i_1<i_2} P(E_{i_1}E_{i_2}) + \cdots \\ & + (-1)^{r+1} \sum_{i_1<\cdots<i_r} P(E_{i_1}\cdots E_{i_r}) + \cdots + (-1)^{n+1} P(E_1 \cdots E_n). \end{aligned}$$

6. **Boole's inequality:** $P\left(\bigcup_i E_i\right) \le \sum_i P(E_i)$. (Truncating inclusion–exclusion after 1, 2, 3, ... terms alternately gives upper, lower, upper, ... bounds.)

Tools: Venn diagrams (fill in from the *middle* outward: start from the triple intersection) and **Venn box** tables:

	A	A^c	
B	$P(A \cap B)$	$P(A^c \cap B)$	$P(B)$
B^c	$P(A \cap B^c)$	$P(A^c \cap B^c)$	$P(B^c)$
	$P(A)$	$P(A^c)$	1

Ross Ex. 4a – Jane's two books

$P(B_1) = .5$, $P(B_2) = .4$, $P(B_1B_2) = .3$. Then $P(B_1 \cup B_2) = .5 + .4 - .3 = .6$, so $P(\text{neither}) = 1 - .6 = .4$. Venn regions: only $B_1 = .2$, both $= .3$, only $B_2 = .1$, neither $= .4$.

2.4 Sample Spaces with Equally Likely Outcomes (Sec. 2.5)

Classical probability

If $S = \{1, \dots, N\}$ is finite and all outcomes are equally likely, then $P(\{i\}) = \frac{1}{N}$ and for any event E

$$P(E) = \frac{\text{number of outcomes in } E}{\text{number of outcomes in } S} = \frac{N(E)}{N(S)}.$$

Classical probability = discrete S + equally likely outcomes; anything else (continuous S, unequal probabilities) is "modern" probability.

Watch out

Rolling 2 dice: the sums $2, \dots, 12$ are **not** equally likely, so $P(8) \ne \frac{1}{11}$. A sum is an *event*, not an outcome. Use the 36 equally likely ordered pairs: $P(\text{sum} = 8) = \frac{5}{36}$.

Order doesn't matter for the answer. When drawing k items at random you may treat the outcome as an *ordered* selection or an *unordered* set – if all outcomes are equally likely in one view, they are in the other (each set corresponds to $k!$ orderings), and you get the same probability. Pick whichever makes counting easier, but **count numerator and denominator the same way**.

Examples from Ross Sec. 2.5

- *(Ex. 5a)* Two dice, sum 7: $\frac{6}{36} = \frac{1}{6}$.
- *(Ex. 5b)* 3 balls drawn from 6 white and 5 black; one white and two black: ordered: $\frac{3 \cdot (6 \cdot 5 \cdot 4)}{11 \cdot 10 \cdot 9} = \frac{360}{990} = \frac{4}{11}$; unordered: $\frac{\binom{6}{1}\binom{5}{2}}{\binom{11}{3}} = \frac{60}{165} = \frac{4}{11}$.
- *(Ex. 5c)* Committee of 5 from 6 men and 9 women; 3 men and 2 women: $\dfrac{\binom{6}{3}\binom{9}{2}}{\binom{15}{5}} = \dfrac{240}{1001}$.
- *(Ex. 5d)* n balls, one special; k drawn: $P(\text{special ball drawn}) = \dfrac{\binom{n-1}{k-1}}{\binom{n}{k}} = \dfrac{k}{n}$.

- *(Ex. 5f)* Poker straight (5 distinct consecutive values, not all one suit): $\dfrac{10\,(4^5-4)}{\binom{52}{5}} \approx .0039$.
- *(Ex. 5g)* Full house: $\dfrac{13\cdot 12\cdot \binom{4}{2}\binom{4}{3}}{\binom{52}{5}} \approx .0014$.
- *(Ex. 5h)* Bridge (52 cards dealt 13 each to 4 players): $P(\text{one player gets all 13 spades}) = 4/\binom{52}{13} \approx 6.3\times 10^{-12}$; $P(\text{each player gets exactly 1 ace}) = \dfrac{4!\binom{48}{12,12,12,12}}{\binom{52}{13,13,13,13}} \approx .1055$.
- *(Ex. 5j)* Shuffled deck, cards turned up until the first ace: each of the 52 cards is equally likely (prob. $\frac{1}{52}$) to be the card that follows the first ace – the ace of spades is no more likely than the two of clubs.

Birthday problem (Ross Ex. 5i; lecture)

n people, ignore leap years, all 365^n birthday assignments equally likely:

$$P(\text{no shared birthday}) = \frac{365\cdot 364\cdots(365-n+1)}{365^n} = \frac{{}_{365}P_n}{365^n}.$$

For $n=25$: $P(\text{at least 2 share}) = 1 - \frac{{}_{365}P_{25}}{365^{25}}$, and $P(\text{exactly one pair shares, nobody else}) = \frac{\binom{25}{2}\cdot {}_{365}P_{24}}{365^{25}}$ (glue the pair into one "person", choose which pair: $\binom{25}{2}$).
Surprise: for $n \geq 23$ the probability of a match exceeds $\frac{1}{2}$ (with $n=50$ it is about .97) – because there are $\binom{23}{2} = 253$ pairs, each with a $\frac{1}{365}$ chance.

Club problem (Ross Ex. 5l; lecture variation)

60 members: 36 tennis (T), 28 squash (S), 18 badminton (B), 22 $T\&S$, 12 $T\&B$, 9 $S\&B$, 4 all three. By inclusion–exclusion, members playing at least one sport: $36+28+18-22-12-9+4 = 43$ (Ross's version used the same counts; "probability" of a random member $= N(C)/N$ turns the counting problem into a probability one). Venn regions: only $T=6$, only $S=1$, only $B=1$, TS only $=18$, TB only $=8$, SB only $=5$, all three $=4$, none $=17$.

- $P((T\cup B)\cap S^c) = \frac{6+8+1}{60} = \frac{1}{4}$.
- $P(T\cup(B\cap S^c)) = \frac{36+1}{60} = \frac{37}{60}$.

Matching problem (Ross Ex. 5m)

N men throw hats into a pile; each randomly takes one. With $E_i =$ "man i gets his own hat": $P(E_{i_1}\cdots E_{i_n}) = \frac{(N-n)!}{N!}$ and there are $\binom{N}{n}$ such terms, so each inclusion–exclusion term equals $\frac{1}{n!}$. Hence

$$P(\text{no matches}) = \sum_{i=0}^{N}\frac{(-1)^i}{i!} \xrightarrow{N\to\infty} e^{-1} \approx .3679.$$

(Not $\to 1$, and not $\to 0$ – a classic surprise.)

3 Chapter 3 – Conditional Probability and Independence

3.1 Conditional Probability (Sec. 3.2)

Conditional probability

For $P(F) > 0$, the probability of E given that F occurred is

$$P(E \mid F) = \frac{P(EF)}{P(F)}.$$

Intuition: F becomes the new (reduced) sample space ("Honey, I shrunk the universe"), and we ask what fraction of it lies in E. When outcomes are equally likely, all outcomes in F stay equally likely, so you can just count inside F. It agrees with the long-run frequency interpretation: among the experiments where F occurs, the fraction in which E also occurs.

Multiplication rule

$$P(EF) = P(F)\,P(E \mid F) = P(E)\,P(F \mid E),$$
$$P(E_1E_2E_3\cdots E_n) = P(E_1)\,P(E_2 \mid E_1)\,P(E_3 \mid E_1E_2)\cdots P(E_n \mid E_1\cdots E_{n-1}).$$

Proof: apply the definition to each factor; the product telescopes.

Examples

- *(lecture / Ross 4a)* Jane's books: $P(B_2 \mid B_1) = \frac{.3}{.5} = .6$.
- *(lecture)* Two dice, first shows 3: $P(\text{sum} = 8 \mid \text{first} = 3) = \dfrac{1/36}{1/6} = \dfrac{1}{6}$ (vs. $\frac{5}{36}$ unconditionally).
- *(Ross Ex. 2a)* Joe's key: 40% left pocket, 40% right pocket. Left search fails: $P(R \mid L^c) = \frac{P(R)}{1-P(L)} = \frac{.4}{.6} = \frac{2}{3}$.
- *(Ross Ex. 2b)* Two flips: $P(\text{both heads} \mid \text{first is heads}) = \frac{1}{2}$ but $P(\text{both heads} \mid \text{at least one head}) = \frac{1}{3}$. (Given at least one head, the three equally likely outcomes are hh, ht, th.)
- *(Ross Ex. 2c)* Bridge: North–South have 8 spades between them; $P(\text{East has exactly 3 of the other 5}) = \dfrac{\binom{5}{3}\binom{21}{10}}{\binom{26}{13}} \approx .339$ (reduced sample space: the 26 East–West cards).
- *(Ross Ex. 2d)* Celine picks French or chemistry by a coin flip; $P(A \mid \text{French}) = \frac{1}{2}$, $P(A \mid \text{chem}) = \frac{2}{3}$: $P(\text{chemistry and A}) = \frac{1}{2}\cdot\frac{2}{3} = \frac{1}{3}$.
- *(Ross Ex. 2e)* 8 red and 4 white balls, two drawn without replacement: $P(R_1R_2) = \frac{8}{12}\cdot\frac{7}{11} = \frac{14}{33}$.
- *(Ross Ex. 2g)* Deck dealt into 4 piles of 13; each pile has exactly one ace: $1\cdot\frac{39}{51}\cdot\frac{26}{50}\cdot\frac{13}{49} \approx .105$.

- *(Ross Ex. 2h)* 8 teams in 4 random pairings; 4 strong teams, none play each other: $\frac{4}{7} \cdot \frac{3}{5} \cdot \frac{2}{3} \cdot 1 = \frac{8}{35}$.

3.2 Bayes's Formula (Sec. 3.3)

Total probability, two-event form (Ross eq. 3.1)

$E = EF \cup EF^c$ (mutually exclusive), so

$$P(E) = P(E \mid F)P(F) + P(E \mid F^c)P(F^c).$$

A weighted average of conditional probabilities, each weighted by the probability of its condition.

Law of Total Probability (Ross eq. 3.4)

If $F_1, \ldots, F_n$ are mutually exclusive and exhaustive ($\bigcup F_i = S$; exactly one occurs), then for any event E

$$P(E) = \sum_{i=1}^{n} P(EF_i) = \sum_{i=1}^{n} P(E \mid F_i)\, P(F_i).$$

Bayes's Formula / Theorem (Ross Prop. 3.1)

With $F_1, \ldots, F_n$ as above and E an event that occurred,

$$P(F_j \mid E) = \frac{P(EF_j)}{P(E)} = \frac{P(E \mid F_j)\, P(F_j)}{\sum_{i=1}^{n} P(E \mid F_i)\, P(F_i)}.$$

The **denominator** collects *all* the ways E can happen; the **numerator** is the one piece you want. It reverses the direction of conditioning: $P(E \mid F_j) \to P(F_j \mid E)$ (effect $\to$ cause). The $P(F_j)$ are prior opinions; the formula shows how evidence E updates them.

Odds and updating (Ross Sec. 3.3)

$\text{odds}(A) = \dfrac{P(A)}{P(A^c)} = \dfrac{P(A)}{1 - P(A)}$. New odds after evidence E:

$$\frac{P(H \mid E)}{P(H^c \mid E)} = \frac{P(H)}{P(H^c)} \cdot \frac{P(E \mid H)}{P(E \mid H^c)}.$$

Evidence supports H only if $P(E \mid H) > P(E \mid H^c)$ – being merely "consistent with" H is not enough (Reese–Schapiro bridge-cheating example).

Recipe: (1) identify the partition F_i (the "causes") and the evidence E; (2) draw a tree and multiply along branches; (3) $P(E)$ = sum of all branches ending in E; (4) $P(F_j \mid E)$ = (branch through F_j)$/P(E)$.

Examples

- *(Ross Ex. 3a)* Accident-prone policyholders: 30% accident prone; yearly accident probability .4 (prone) or .2 (not). $P(A) = .4(.3) + .2(.7) = .26$; $P(\text{prone} \mid A) = \frac{.12}{.26} = \frac{6}{13}$.
- *(Ross Ex. 3c)* Multiple choice with m options; knows answer with probability p, otherwise guesses: $P(\text{knew} \mid \text{correct}) = \dfrac{p}{p + \frac{1}{m}(1-p)} = \dfrac{mp}{1 + (m-1)p}$; for $m = 5, p = \frac{1}{2}$ it's $\frac{5}{6}$.
- *(Ross Ex. 3d)* Blood test: 95% detection, 1% false positives, 0.5% prevalence. $P(D \mid \text{positive}) = \dfrac{.95(.005)}{.95(.005) + .01(.995)} = \dfrac{95}{294} \approx .323$. So $P(\text{healthy} \mid \text{positive}) = 1 - .323 = .677$. *Sanity check:* per 200 people tested, about .95 true positives vs. $199(.01) \approx 1.99$ false positives.
- *(Ross Ex. 3e)* Doctor, $P(D) = .6$, test positive always if diseased, 30% of the time if healthy diabetic: $P(D \mid +) = \frac{.6}{.6 + .3(.4)} \approx .833$.
- *(Ross Ex. 3f)* Inspector: 60% sure of guilt; trait found with 20% frequency in population: $P(G \mid C) = \frac{.6}{.6 + .2(.4)} \approx .882$.
- *(Ross Ex. 3i)* Urn with two type-A coins (heads prob. $\frac{1}{4}$) and one type-B coin (heads prob. $\frac{3}{4}$); a random coin lands heads: $P(A \mid H) = \dfrac{\frac{2}{3} \cdot \frac{1}{4}}{\frac{2}{3} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{3}{4}} = \frac{2}{5}$ (odds $\frac{2}{3} : 1$).
- *(Ross Ex. 3l)* Three cards: red–red, black–black, red–black. A random card shows red; $P(\text{other side black}) = \frac{1}{3}$ (**not** $\frac{1}{2}$): there are 6 equally likely card-sides, 3 red, of which only 1 has a black back.
- *(Ross Ex. 3n)* Flashlights: types 1, 2, 3 are 20%, 30%, 50% of the bin and last > 100 hours with probability $.7, .4, .3$. $P(A) = .14 + .12 + .15 = .41$; $P(\text{type 1} \mid A) = \frac{14}{41}$, $P(\text{type 2} \mid A) = \frac{12}{41}$, $P(\text{type 3} \mid A) = \frac{15}{41}$.
- *(lecture / SOA)* ER patients: critical/serious/stable = 10/30/60%, death rates 40/10/1%. $P(\text{serious} \mid \text{survived}) = \frac{.3(.9)}{.1(.6) + .3(.9) + .6(.99)} = \frac{.27}{.924} \approx .292$ (answer (B)).
- *(lecture)* Urn without replacement (3 white, 2 red): $P(W_2) = \frac{3}{5} \cdot \frac{2}{4} + \frac{2}{5} \cdot \frac{3}{4} = \frac{3}{5}$; $P(W_1 \mid R_2) = \frac{6/20}{8/20} = \frac{3}{4}$. Two dice, sum is 8: $P(\text{first} = 3 \mid \text{sum} = 8) = \frac{1}{5}$, $P(\text{both} = 3 \mid 8) = 0$, $P(\text{at least one is } 3 \mid 8) = \frac{2}{5}$.

Two-children problem (Ross Ex. 3m)

"A mother with two children is seen with a girl; what is the probability both are girls?" is *not well-defined* until you specify how the child seen was chosen: it can be $\frac{1}{2}$ or $\frac{1}{3}$ (or $\frac{1}{1+2q}$) depending on the assumption. Always state the sample space.

3.3 Independent Events (Sec. 3.4)

Independence of two events

E and F are **independent** if

$$P(EF) = P(E)\,P(F),$$

equivalently $P(E \mid F) = P(E)$, equivalently $P(F \mid E) = P(F)$ (if one holds, all hold): knowing one occurred doesn't change the probability of the other. Events that are not independent are **dependent**.

Complements (Ross Prop. 4.1)

If E and F are independent, so are E and F^c (hence also E^c and F, and E^c and F^c).
Proof: $P(EF^c) = P(E) - P(EF) = P(E) - P(E)P(F) = P(E)\big(1 - P(F)\big) = P(E)P(F^c)$.

Independence of 3 or more events

Events $E_1, \ldots, E_n$ are independent if *every* subcollection $E_{i_1}, \ldots, E_{i_r}$ $(r \le n)$ satisfies $P(E_{i_1} \cdots E_{i_r}) = P(E_{i_1}) \cdots P(E_{i_r})$ – all pairs, all triples, all quadruples, ... For three events E, F, G you must verify all four:

$$P(EF) = P(E)P(F), \qquad P(EG) = P(E)P(G),$$
$$P(FG) = P(F)P(G), \qquad P(EFG) = P(E)P(F)P(G).$$

If E, F, G are independent then E is independent of any event built from F and G (e.g. $F \cup G$). An infinite set of events is independent if every finite subset is. **Independent trials:** subexperiments whose outcomes don't affect each other.

Watch out

Pairwise independence does not imply independence. Ross Ex. 4e: two dice; E = sum is 7, F = first die is 4, G = second die is 3. E is independent of F and of G separately, but $P(E \mid FG) = 1 \neq P(E)$. Lecture example: toss a fair coin 3 times; A = first two match, B = last two match, C = first and third match; each pair is independent ($P = \frac{1}{4}$) but $P(ABC) = \frac{1}{4} \neq \frac{1}{8}$. Also: mutually exclusive events with positive probability are *never* independent.

Examples

- *(Ross Ex. 4a)* Draw a card: E = ace, F = spade are independent ($\frac{1}{52} = \frac{4}{52} \cdot \frac{13}{52}$).
- *(Ross Ex. 4b)* Two flips: E = first is heads, F = second is tails: independent ($\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$).
- *(Ross Ex. 4c / lecture)* Two dice, F = first die is 4: sum = 6 is **not** independent of F ($\frac{1}{36} \neq \frac{5}{36} \cdot \frac{1}{6}$); sum = 7 **is** independent ($\frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6}$); sum = 8 with first die 3: not independent.
- *(lecture)* 50% of clients married, 30% have a degree, 15% both: .5·.3 = .15, so independent.
- *(Ross Ex. 4f)* Independent trials, success probability p: $P(\text{at least one success in } n) = 1 - (1-p)^n$; $P(\text{exactly } k \text{ successes in } n) = \binom{n}{k} p^k (1-p)^{n-k}$; $P(\text{all } n \text{ succeed}) = p^n$.

- *(Ross Ex. 4g)* Parallel system (works if at least one component works), independent components with success probabilities p_i: $P(\text{works}) = 1 - \prod_{i=1}^{n}(1-p_i)$.
- *(Ross Ex. 4h)* Repeated rolls of two dice: $P(\text{a 5 appears before a 7}) = \dfrac{4/36}{4/36+6/36} = \dfrac{2}{5}$. In general, for mutually exclusive E, F and independent trials, $P(E \text{ before } F) = \dfrac{P(E)}{P(E)+P(F)}$.

Recap: $P(A \text{ or } B) = P(A) + P(B) - P(AB)$ in general ($= P(A) + P(B)$ if mutually exclusive). $P(A \text{ and } B) = P(A)P(B \mid A)$ in general ($= P(A)P(B)$ if independent).

4 Chapter 4 – Discrete Random Variables (4.1–4.5)

4.1 Random Variables, PMF, CDF (Sec. 4.1)

Random variable

A **random variable** (R.V.) is a real-valued function defined on the sample space – a numerical quantity determined by the outcome of the experiment (e.g. the number of heads in 3 flips, the sum of two dice).

Lecture definitions: *preliminary:* the range of a numerically valued function whose domain is the sample space. *Preferred:* the numerically valued domain of a function whose range is a probability distribution – i.e. we can forget the sample space and specify only the values and their probabilities (different experiments can give the same R.V.; we can even "invent" one by making up numbers satisfying the conditions below).

Notation: capital X for the R.V., lowercase x for a value it takes.

Probability mass function (PMF) / probability distribution

For a discrete R.V. X: $p(a) = P(X = a)$. It must satisfy (1) $p(x) \geq 0$, and (2) $\sum_x p(x) = 1$. A named R.V. has one PMF; a different distribution means a different R.V.

Cumulative distribution function (CDF)

$F(x) = P(X \leq x)$ for $-\infty < x < \infty$. It is **nondecreasing** (if $a < b$ then $\{X \leq a\} \subseteq \{X \leq b\}$, so $F(a) \leq F(b)$). For a discrete R.V.

$$F(a) = \sum_{x \leq a} p(x), \qquad p(x_i) = F(x_i) - F(x_{i-1}),$$

so F is a **step function** whose jump at x_i has size $p(x_i)$. Often $P(X > a) = 1 - F(a)$ is the easy way to compute a tail probability. (Lecture: the CDF can be used to simulate any R.V. from a uniform random number.)

Examples

- *(lecture)* X = heads in 2 fair tosses:

x	0	1	2
$p(x)$	.25	.50	.25
$F(x)$	.25	.75	1.00

- *(Ross Ex. 1a)* Y = heads in 3 fair flips: $p(0) = \frac{1}{8}$, $p(1) = \frac{3}{8}$, $p(2) = \frac{3}{8}$, $p(3) = \frac{1}{8}$.
- *(Ross Ex. 1b)* Two clients, deaths independent with probabilities .05 (younger) and .10 (older); X = payout in units of \$100,000: $P(X = 0) = .95(.9) = .855$; $P(X = 1) = .05(.9) + .95(.1) = .140$; $P(X = 2) = .005$.
- *(Ross Ex. 1c)* Choose 4 of balls numbered 1–20; X = largest number: $P(X = i) = \frac{\binom{i-1}{3}}{\binom{20}{4}}$, $i = 4, \ldots, 20$; and $P(X \le 10) = \frac{\binom{10}{4}}{\binom{20}{4}}$ (all four in $\{1, \ldots, 10\}$).
- *(Ross Ex. 1d)* Flip until a head or n flips: $P(X = k) = (1-p)^{k-1}p$ for $k < n$, $P(X = n) = (1-p)^{n-1}$.

4.2 Discrete Random Variables (Sec. 4.2)

Discrete R.V.

A R.V. that can take on at most a countable number of values (finite, or countably infinite such as $1, 2, 3, \ldots$).

Examples

- *(lecture)* Toss a fair coin until the first head; X = number of tosses: $p(n) = \left(\frac{1}{2}\right)^{n-1}\frac{1}{2} = \frac{1}{2^n}$. Total $= \frac{1/2}{1-1/2} = 1$ (geometric series $\frac{a}{1-r}$).
- *(Ross Ex. 2a)* $p(i) = c\lambda^i/i!$, $i = 0, 1, 2, \ldots$ Since $\sum \lambda^i/i! = e^\lambda$, $c = e^{-\lambda}$; then $P(X = 0) = e^{-\lambda}$ and $P(X > 2) = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2}$.
- *(Ross)* CDF of $p(1) = \frac{1}{4}, p(2) = \frac{1}{2}, p(3) = \frac{1}{8}, p(4) = \frac{1}{8}$: $F(a) = 0$ for $a < 1$; $\frac{1}{4}$ for $1 \le a < 2$; $\frac{3}{4}$ for $2 \le a < 3$; $\frac{7}{8}$ for $3 \le a < 4$; 1 for $a \ge 4$.
- *(Review #8)* $P(X = x) = cx^2$ for $x = 1, 2, 3$: $c(1 + 4 + 9) = 1 \Rightarrow c = \frac{1}{14}$.

4.3 Expected Value (Sec. 4.3)

Summation rules (lecture): $\sum_{i=1}^{n} c = cn$; $\sum cx_i = c\sum x_i$; $\sum(x_i \pm y_i) = \sum x_i \pm \sum y_i$. They do *not* work for products/quotients: $\sum x_i y_i \neq \sum x_i \sum y_i$. A **weighted average** is $\frac{\sum x_i w_i}{\sum w_i}$.

Expected value

For a discrete R.V. X with PMF p:

$$\mathbb{E}[X] = \sum_{x:\, p(x)>0} x\, p(x).$$

A weighted average of the possible values, weighted by their probabilities (which sum to 1). It plays the role of the mean μ in statistics, is the long-run average per trial (frequency interpretation), and is the "center of gravity" of the probability masses.

Examples

- *(lecture)* Heads in 2 tosses: $\mathbb{E}[X] = 0(.25) + 1(.5) + 2(.25) = 1$.
- *(Ross Ex. 3a)* Fair die: $\mathbb{E}[X] = \frac{1}{6}(1 + \cdots + 6) = \frac{7}{2}$.
- *(Ross Ex. 3b)* **Indicator variable** $I_A = 1$ if A occurs, 0 otherwise: $\mathbb{E}[I_A] = P(A)$.
- *(Ross Ex. 3d / lecture)* 120 students on buses of 36, 40, 44; pick a student at random, X = number on that student's bus: $p = \frac{36}{120}, \frac{40}{120}, \frac{44}{120}$, so $\mathbb{E}[X] = \frac{1208}{30} \approx 40.27$, larger than the average bus size 40 (big buses are more likely to contain the chosen student).
- *(Ross Ex. 3c)* Quiz show with two questions worth V_1, V_2 known with probabilities P_1, P_2: attempt question 1 first iff $\frac{V_1 P_1}{1-P_1} > \frac{V_2 P_2}{1-P_2}$.

4.4 Expectation of a Function of a R.V. (Sec. 4.4)

Proposition 4.1 and linearity rules

1. $\mathbb{E}[g(X)] = \sum_i g(x_i)\, p(x_i)$ (use the PMF of X; no need to find the PMF of $g(X)$). *Proof:* group the terms with equal $g(x_i) = y_j$: $\sum_j y_j P(g(X) = y_j)$.
2. $\mathbb{E}[c] = c$. 3. $\mathbb{E}[cX] = c\,\mathbb{E}[X]$.
3. **Corollary 4.1:** $\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b$. *Proof:* $\sum(ax_i + b)p(x_i) = a\sum x_i p(x_i) + b\sum p(x_i) = a\mathbb{E}[X] + b \cdot 1$.
4. $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ (later, with joint distributions).

Rules 2–3 and Cor. 4.1 are the cases where g is linear; then $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$. The nth **moment** of X is $\mathbb{E}[X^n] = \sum x^n p(x)$; $\mathbb{E}[X]$ is the mean or first moment.

Watch out

In general $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$. E.g. $\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2$.

Examples

- *(Ross Ex. 4a)* $P(X=-1)=.2, P(X=0)=.5, P(X=1)=.3$. $\mathbb{E}[X^2]=(-1)^2(.2)+0+1^2(.3)=.5$, whereas $(\mathbb{E}[X])^2=(.1)^2=.01$.
- *(lecture)* $X=$ heads in 2 tosses, $Y=2X+3$, $Z=X^2$: $\mathbb{E}[Y]=2(1)+3=5$ (shortcut) and $\mathbb{E}[Z]=0(.25)+1(.5)+4(.25)=1.5\neq 1=(\mathbb{E}[X])^2$.

4.5 Variance (Sec. 4.5)

Variance and standard deviation

Variance measures spread about the mean $\mu=\mathbb{E}[X]$:

$$\mathrm{Var}[X]=\sigma_X^2=\mathbb{E}\big[(X-\mu)^2\big]=\sum_i (x_i-\mu)^2 p(x_i), \qquad \mathrm{SD}[X]=\sqrt{\mathrm{Var}[X]}.$$

(The average signed deviation $\mathbb{E}[X-\mu]$ is always 0; $\mathbb{E}|X-\mu|$ works but is awkward algebraically; squaring is convenient.) Example: $W\equiv 0$, $Y=\pm 1$ with prob. $\frac{1}{2}$ each, $Z=\pm 100$ with prob. $\frac{1}{2}$ each all have mean 0 but increasing spread.

Computational formula

$$\mathrm{Var}[X]=\mathbb{E}[X^2]-\big(\mathbb{E}[X]\big)^2=\mathbb{E}[X^2]-\mu^2.$$

Since $\mathrm{Var}[X]\geq 0$ (a sum of nonnegative terms), $\mathbb{E}[X^2]\geq(\mathbb{E}[X])^2$ always.

Proof: $\sum(x_i-\mu)^2p(x_i)=\sum x_i^2p(x_i)-2\mu\sum x_ip(x_i)+\mu^2\sum p(x_i)=\mathbb{E}[X^2]-2\mu^2+\mu^2=\mathbb{E}[X^2]-\mu^2$.

Rules for variance

1. $\mathrm{Var}[g(X)]=\sum\big(g(x_i)-\mathbb{E}[g(X)]\big)^2p(x_i)$.
2. $\mathrm{Var}[c]=0$ (no dispersion).
3. $\mathrm{Var}[aX]=a^2\,\mathrm{Var}[X]$.
4. $\mathrm{Var}[aX+b]=a^2\,\mathrm{Var}[X]$; in particular $\mathrm{Var}[X+b]=\mathrm{Var}[X]$ (shifting doesn't change spread).
5. $\mathrm{Var}[X+Y]=\mathrm{Var}[X]+\mathrm{Var}[Y]$ **only if X and Y are independent** (later).

Examples

- *(lecture)* Heads in 2 tosses: $\mathbb{E}[X^2]=0+.5+1.0=1.5$, so $\mathrm{Var}[X]=1.5-1^2=0.5$, $\mathrm{SD}=\sqrt{.5}$.
- *(Ross Ex. 5a)* Fair die: $\mathbb{E}[X]=\frac{7}{2}$, $\mathbb{E}[X^2]=\frac{91}{6}$, so $\mathrm{Var}[X]=\frac{91}{6}-\frac{49}{4}=\frac{35}{12}$.
- *(lecture)* Data set $V=\{1,1,2,2,5,8,8,9,9\}$ (mean 5): $\mathbb{E}[V^2]=\frac{325}{9}$, $\mathrm{Var}[V]=\frac{325}{9}-25=\frac{100}{9}\approx 11.11$. $N=\{4,4,4,5,5,5,6,6,6\}$: $\mathrm{Var}[N]=\frac{2}{3}$.

5 Review-Sheet Checklist (state, formula, explain, example)

The professor will ask you to **state a formula, define, and perhaps explain in plain language or give an example** for a subset of these. Cover the right-hand side and test yourself.

Concept	**What you must be able to say**
(a) Multiplication principle	Stages with $n_1, \dots, n_r$ outcomes each (regardless of earlier results) $\Rightarrow$ $n_1 n_2 \cdots n_r$ total outcomes.
(b) Permutation	Ordered selection; ${}_nP_r = \frac{n!}{(n-r)!}$; $n!$ orderings of n objects; with alike objects divide by $n_i!$.
(c) Combination	Unordered selection of r from n; $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.
(d) Partition	Splitting n distinct objects into groups of sizes $r_1, \dots, r_k$ ($\sum r_i = n$): $\frac{n!}{r_1! \cdots r_k!}$.
(e) Binomial theorem	$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$.
(g) Sample space	Set of all possible outcomes of an experiment.
(h) Event	A subset of the sample space.
(i) Mutually exclusive	$A \cap B = \emptyset$; cannot occur together.
(j) 3 axioms	$0 \le P(E) \le 1$; $P(S) = 1$; $P(\cup E_i) = \sum P(E_i)$ for mutually exclusive E_i.
(k) Equally likely	$P(E) = N(E)/N(S)$, valid only when outcomes are equally likely.
(l) Conditional probability	$P(B \mid A) = P(A \cap B)/P(A)$, $P(A) > 0$.
(m) Bayes' formula	$P(F_j \mid E) = \frac{P(E\|F_j)P(F_j)}{\sum_i P(E\|F_i)P(F_i)}$, F_i mutually exclusive and exhaustive.
(n) Independence (2 events)	$P(A \cap B) = P(A)P(B)$ (equivalently $P(A \mid B) = P(A)$).
(o) Independence (3+ events)	Every subcollection's intersection probability is the product – all pairs, all triples, ...; pairwise is not enough.
(p) Random variable	Numerically valued variable whose values occur with given probabilities (function from sample space to numbers).
(q) PMF	$p(x) = P(X = x)$; $p \ge 0$ and $\sum p = 1$.
(r) CDF	$F(x) = P(X \le x)$; nondecreasing; step function for discrete X.
(s) Expected value	$\mathbb{E}[X] = \sum x_i p(x_i)$, probability-weighted average.

(Item (f), proof by induction, is skipped.)

6 Review Sheet Practice Problems – Worked Solutions

Try each problem yourself before reading the answer.

2. Phone numbers. $8 \cdot 10^6$ numbers with first digit in $\{2, \dots, 9\}$; remove 555-xxxx: $\boxed{7{,}990{,}000}$.

3. **Blocks.** $\dfrac{12!}{4!\,6!\,1!\,1!} = \boxed{27{,}720}$.

4. **Heads = dots.** The die shows ≥ 1, so heads must be 1 or 2. $P(1 \text{ head}) = \frac{1}{2}$, $P(2 \text{ heads}) = \frac{1}{4}$. Total $= \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{1}{6} = \boxed{\frac{1}{8}}$.

5. **Socks.** $\dfrac{\binom{4}{2} + \binom{8}{2}}{\binom{12}{2}} = \dfrac{6 + 28}{66} = \boxed{\frac{17}{33}}$.

6. **Insurance** (82 clients, 62 auto, 37 home, assuming every client has at least one).

   (a) $N(\text{both}) = 62 + 37 - 82 = 17$, so $P = \boxed{\frac{17}{82} \approx 0.207}$.

   (b) $P(\text{auto})P(\text{home}) = \frac{62}{82} \cdot \frac{37}{82} \approx 0.341 \neq 0.207$, so **not independent**.

7. **Blood types.** $P(\text{test} = A) = .41(.88) + .09(.04) + .04(.10) + .46(.04) = .3868$. Bayes: $P(\text{type } A \mid \text{test } A) = \dfrac{.3608}{.3868} \approx \boxed{0.933}$.

8. **PMF constant.** $c(1 + 4 + 9) = 1 \Rightarrow c = \boxed{\frac{1}{14}}$.

7 Textbook Self-Test Problems – What's In Scope

The professor said to expect **1–2 problems similar to the Self-Test Problems and Exercises** (third problem group at the end of each chapter; full solutions are in the back of the book). Page numbers are the *printed* pages of the 10th edition (the PDF viewer's page numbers differ).

Chapter	In-scope problems (topic)
Ch. 1 (p. 20)	1 (letter arrangements with adjacency/order conditions); 2 (seating by nationality); 3 (officers with restrictions); 4 (7 of 10 questions); 5 (dividing gifts – partition); 6 (license plates); 7, 17 (combinatorial identities); 8 (n-digit numbers); 9 (3 students from 3 classes); 10 (5-digit numbers, each digit at most twice); 11 (committee with no married couple); 12 (at least 3 women and 2 men); 15 (posted exam results); 16 (subsets containing 1–5); 18 (parent–child pairs); 19 (plates, no repeats). *Skip:* 13, 14 (starred, Sec. 1.6).
Ch. 2 (p. 56)	1 (sample space of a meal); 2 (suit/shirt/tie inclusion–exclusion); 3 (14th card an ace); 4 (max/min of two events); 5 (top four cards); 6 (two urns, same color); 7 (lottery); 8 (committee by class); 9 (inclusion–exclusion for counts); 10 (horse race); 11 (all four suits in a 5-card hand); 12 (roommates); 13 (letters from two words); 14, 15 (Boole's inequality and its corollary). *Skip:* 16 (partitions of a set into subsets), 20–21 (infinite/runs).
Ch. 3 (p. 116)	3.1 (bridge, aces); 3.2 (battery life, conditional); 3.3 (maximize white-ball probability); 3.4 (urn transfer, Bayes); 3.5 (urn, $P(R_i)$); 3.6 (Polya urn); 3.7 (which card is an ace); 3.8 (odds updating); 3.9 (plant and neighbor, Bayes); 3.10 (red/green/blue balls); 3.11 (battery types, Bayes); 3.12 (two books); 3.13 (color depletion order); 3.14 (message passing); 3.15 (rat genetics, Bayes); 3.16–3.17 (relay/k-out-of-n systems, independence); 3.18 (roulette fallacy).
Ch. 4 (p. 186)	Only 4.1–4.6: 4.1 (find $\mathbb{E}[X]$ from constraints); 4.2 (ratio condition); 4.3 (expected flips); 4.4 (family-size paradox, like the bus example); 4.5 (find $P(X=0)$ from $\mathbb{E}[X] = 3\,\mathrm{Var}[X]$ – use $\mathrm{Var} = \mathbb{E}[X^2] - (\mathbb{E}X)^2$ for a 0/1 variable); 4.6 (value of information, expected payoff). 4.7 (envelope strategy) uses only expectation and is optional. *Skip* 4.8 onward (binomial, Poisson, geometric, . . .).

Ch. 4 hint for 4.5: if $X \in \{0,1\}$ with $p = P(X=1)$, then $\mathbb{E}[X] = p$, $\mathbb{E}[X^2] = p$, $\mathrm{Var}[X] = p(1-p)$.

Problem-solving checklist

1. **Name the experiment and the sample space.** Are outcomes equally likely? If not, don't use $N(E)/N(S)$.
2. **Counting?** Ordered vs. unordered; with vs. without replacement; labeled vs. unlabeled groups; subtract forbidden cases.
3. **"At least one"** → complement. **"A or B"** → inclusion–exclusion. **"A and B"** → multiplication rule / independence.
4. **Information received / reversed conditioning** → conditional probability / Bayes; draw the tree and sum *all* branches in the denominator.
5. **Random variables:** list values, PMF (check it sums to 1), then $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\mathrm{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$; use the linear shortcuts for $aX + b$.
6. **Check:** probabilities in $[0,1]$, PMF sums to 1, answer passes a sanity check.

8 Quick-Reference Formula Sheet

Counting

$n_1 n_2 \cdots n_r$
${}_nP_r = \frac{n!}{(n-r)!}$
$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \binom{n}{n-r}$
$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$
$\frac{n!}{n_1!\cdots n_k!}$ (partitions / repeated letters)
$(x+y)^n = \sum \binom{n}{k} x^k y^{n-k}$

Probability basics

$P(E^c) = 1 - P(E)$
$P(E \cup F) = P(E) + P(F) - P(EF)$
$P(\cup E_i) \le \sum P(E_i)$
$P(E) = N(E)/N(S)$ (equally likely)

Conditional / independence

$P(E \mid F) = P(EF)/P(F)$
$P(EF) = P(F)P(E \mid F)$
$P(E) = \sum P(E \mid F_i)P(F_i)$
Bayes: $\frac{P(E|F_j)P(F_j)}{P(E)}$
Odds: new = old $\times \frac{P(E|H)}{P(E|H^c)}$
Indep.: $P(EF) = P(E)P(F)$

Random variables

$p(x) \ge 0$, $\sum p(x) = 1$; $F(x) = P(X \le x)$
$\mathbb{E}[X] = \sum x\, p(x)$; $\mathbb{E}[g(X)] = \sum g(x)p(x)$
$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$
$\mathrm{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$
$\mathrm{Var}[aX + b] = a^2 \mathrm{Var}[X]$; SD $= \sqrt{\mathrm{Var}}$